

Figure 5.3 XQ-58A Valkyrie (2019).⏎

Technology (LCAAT) program. The image highlights its blended-wing, low-observable design, optimized for stealth and high-speed performance. The angular fuselage and flush air intakes contribute to radar signature reduction, while the tricycle landing gear supports independent ground operations. This UAV is designed for manned–unmanned teaming, operating alongside fighter jets like the F-35 and F-22 to provide reconnaissance, electronic warfare, and strike support. With AI-driven autonomy, the Valkyrie executes coordinated missions, adapts to threats in real time, and represents a major leap toward intelligent, cost-effective aerial warfare systems in the era of fourth-generation UAVs.

5.6.2.2 CH-7 Rainbow (2019)

The CH-7 Rainbow, developed by the China Aerospace Science and Technology Corporation (CASC), is a cutting-edge stealth unmanned combat aerial vehicle (UCAV) that exemplifies China’s rapid advancement in fourth-generation drone technology. First unveiled in 2018 at Zhuhai Airshow and fully operational by 2019, the CH-7 is designed for strategic